

Solution: Both $f \circ g$ and $g \circ f$ are defined.

$$f \circ g = f(g(x)) = f(3x+2) = 2(3x+2)+3 = 6x+7$$

$$g \circ f = g(f(x)) = g(2x+3) = 3(2x+3)+2 = 6x+11 \quad \square$$

Example 25: Let f and g be the functions defined by $f: \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$ with $f(x) = x^2$ and $g: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}$ with $g(x) = \sqrt{x}$ (where $\sqrt{x}$ is the nonnegative sq. root of x). What is the function $(f \circ g)(x)$?

Solution: The domain of $f(g(x))$ is the domain of g which is $\mathbb{R}^+ \cup \{0\}$.

The range of $g(x)$ is a subset of $\mathbb{R}$ which is the domain of f .

$$f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 = x \text{ for all } x \in \mathbb{R}^+ \cup \{0\}$$

When the composition of a function and its inverse is formed, in either order, an identity function is obtained. To see this, suppose that f is a one-to-one correspondence from the set A to the set B . Then the inverse function f^{-1} exists and is a one-to-one correspondence from B to A . The inverse function reverses the correspondence of the original function, so $f^{-1}(b) = a$ when $f(a) = b$, and $f(a) = b$ when $f^{-1}(b) = a$. Hence,

$$(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a \text{ and } (f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b.$$

Consequently $f^{-1} \circ f = t_A$ and $f \circ f^{-1} = t_B$, where t_A and t_B are the identity functions on the sets A and B respectively. That is $(f^{-1})^{-1} = f$.

2.3.4 The Graphs of Functions

Definition 11: Let f be a function from the set A to the set B . The **graph** of the function f is the set of ordered pairs $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.

Example 26: Display the graph of the function $f(x) = 2x+1$ from the set of integers to the set of integers.

Solution:

